Unit: Unit 3: Functions
Topic: Composite Functions

Name: ____________________
Date: ______________

Advanced (Habanero)

Q1. Evaluate $f(g(x))$ where $f(x) = x^2$ and $g(x) = 2x - 1$.
(A) $4x^2 - 4x + 1$
(B) $4x^2 - 2x + 1$
(C) $2x^2 - 2x + 1$
(D) $2x^2 + 2x + 1$
(E) $x^2 + 2x - 1$

Q2. Determine the domain of $f(g(x))$ where $f(x) = \frac{1}{x}$ and $g(x) = x^2 - 4$.
(A) $(-\infty, -2) \cup (2, \infty)$
(B) $(-\infty, -2] \cup [2, \infty)$
(C) $(-2, 2)$
(D) $[-2, 2]$
(E) $(-\infty, \infty)$

Q3. Given $f(x) = x + 2$ and $g(x) = 2x - 3$, what is $f(g(2))$?
(A) 3
(B) 5
(C) 7
(D) 1
(E) 0

Q4. If $f(x) = \sqrt{x}$ and $g(x) = x^2 + 1$, what is the domain of $f(g(x))$?
(A) $[-1, \infty)$
(B) $[0, \infty)$
(C) $(-\infty, \infty)$
(D) $[1, \infty)$
(E) $(-1, 1)$

Q5. For $f(x) = x^3$ and $g(x) = x - 2$, what is $f(g(3))$?
(A) 1
(B) 8
(C) 27
(D) 0
(E) -1

Q6. Determine the range of $f(g(x))$ where $f(x) = x^2$ and $g(x) = x - 3$.
(A) $[0, \infty)$
(B) $[-\infty, -1]$
(C) $[1, \infty)$
(D) $[-1, 1]$
(E) $(-\infty, \infty)$

Q7. Given $f(x) = \frac{1}{x+1}$ and $g(x) = 2x$, what is $f(g(1))$?
(A) $\frac{1}{3}$
(B) $\frac{1}{2}$
(C) $\frac{2}{3}$
(D) $\frac{3}{4}$
(E) 1

Q8. If $f(x) = x^2 - 4$ and $g(x) = x + 1$, what is the domain of $f(g(x))$?
(A) $[-2, 2]$
(B) $(-\infty, -3] \cup [1, \infty)$
(C) $(-3, 1)$
(D) $(-\infty, \infty)$
(E) $[-1, 1]$

Open Ended Questions (FRQ)

Q9. A water tank can be filled at a rate given by $f(t) = 2t$ gallons per minute, and the water is then processed at a rate given by $g(x) = \frac{1}{x}$. Determine $f(g(x))$ and describe its practical meaning in this context.

Q10. For the functions $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 2}$, find the domain and range of $f(g(x))$ and provide a real-world scenario where such composite function could be applied.

Chef's Tips

- Ensure students understand the concept of composite functions before attempting the test.
- Provide examples of real-world applications to help students interpret compositions.
- Allow the use of calculators for calculations but ensure understanding of concepts.